CS 486/686
Neural Networks and Learned Embeddings

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Lecture 18

From fixed features to learned vector representations

Recorded lecture (asynchronous)

> 🎥 This lecture is **pre-recorded** - I'm away at **ICML** this week, so there's no in-person class today. Watch at your own pace.
>
> 📅 **Due today (Thu Jul 9):** CS686 project proposal. **Assignment 2** is now due **Thu Jul 16**.
>
> 💬 Questions? Post on **Piazza** - the TAs are available, and I'll follow up when I'm back.

What you should take away

One thread runs through the whole lecture: useful vectors can be **learned**.

build

Explain why hidden layers need nonlinearities.

represent

Explain embeddings as learned vectors, including contextual ones.

inspect

Use projection, neighbors, and clusters to reason about embedding spaces.

The later examples are applications of the same idea: learned spaces become interfaces.

Act 1: learn the features

L17 trained models on fixed input vectors.

Today, the model also learns the vector representation.

The problem with fixed features

L17 assumed the useful input vector already exists.

pixels

Where is the cat ear?

word IDs

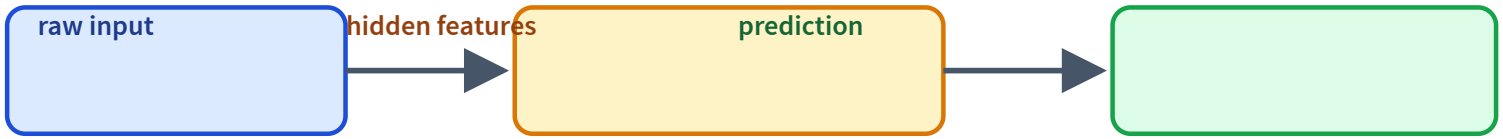
Why is dog
close to puppy?

screenshots

Which region
is a button?

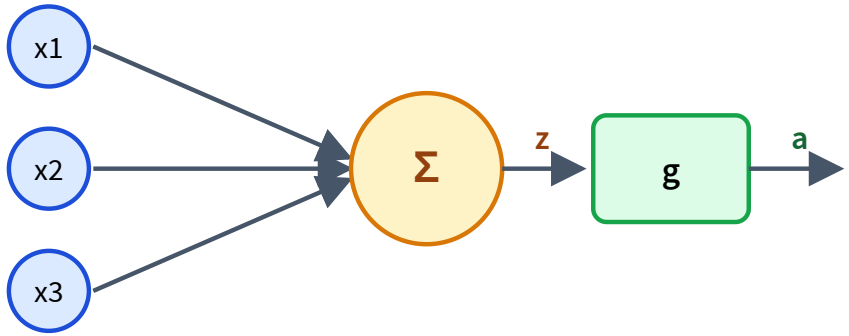
Modern ML learns the features too.

Neural nets learn representations



The hidden layer is not just computation: it is a learned view of the input.

A neuron is a feature detector



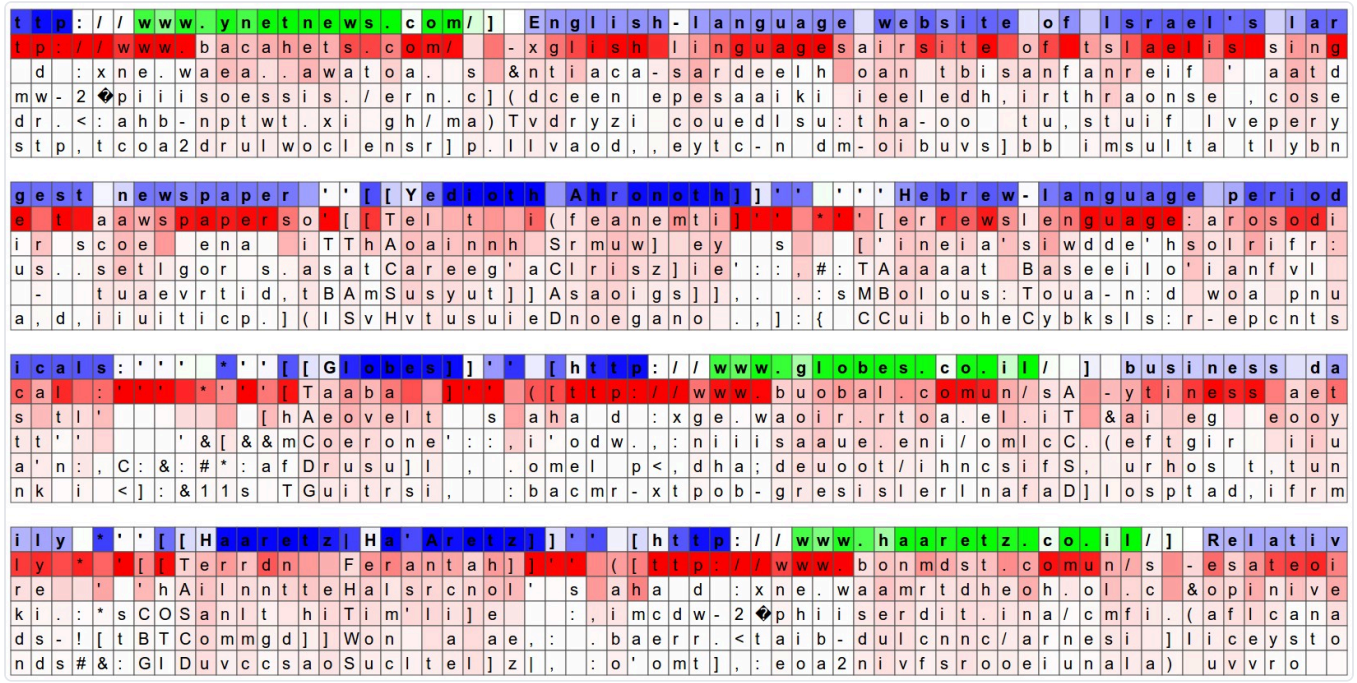
inputs x_1, x_2, x_3

weighted sum: $z = w^\top x + b$

activation: $a = g(z)$

A real learned neuron

Each character is colored by **one** neuron's activation in a character-level RNN.

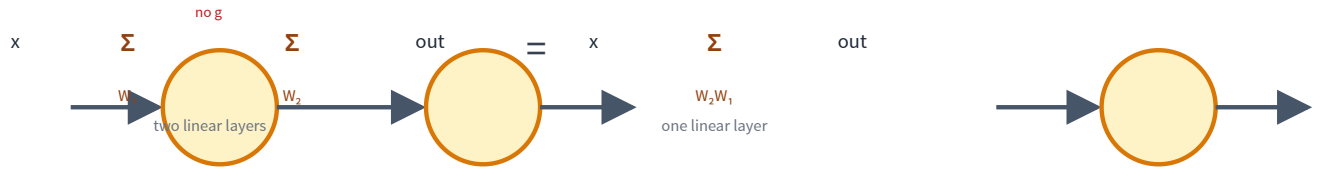


Nobody programmed it — this neuron learned to fire inside URLs.

Karpathy, "The Unreasonable Effectiveness of Recurrent Neural Networks," 2015.

Why add a nonlinearity?

Two linear layers in a row collapse into one:

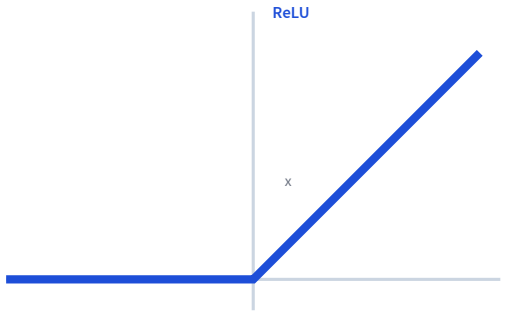


$$W_2(W_1 x) = (W_2 W_1)\, x$$

Depth alone adds nothing — we need a nonlinearity between the layers.

The fix: a nonlinearity

Insert g between layers: $h = g(W_1 x), \quad \hat{y} = W_2 h.$



$$g(x) = \max(0, x)$$

The bend is what lets stacked layers represent nonlinear functions.

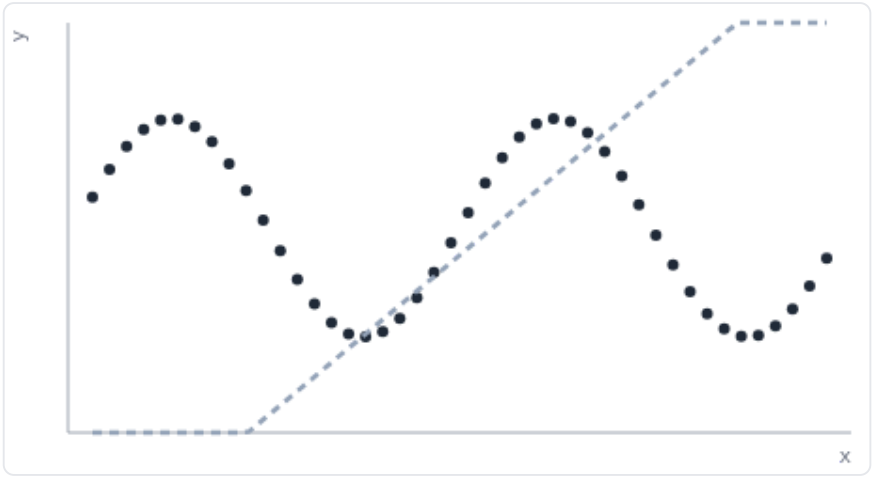
Playground: does nonlinearity matter? LIVE

Same net, activation off vs. on — try it on the live slides.

`nn.Sequential(nn.Linear(1,12), nn.Tanh(), nn.Linear(12,1))` · target: y = sin(2x)

activation: OFF

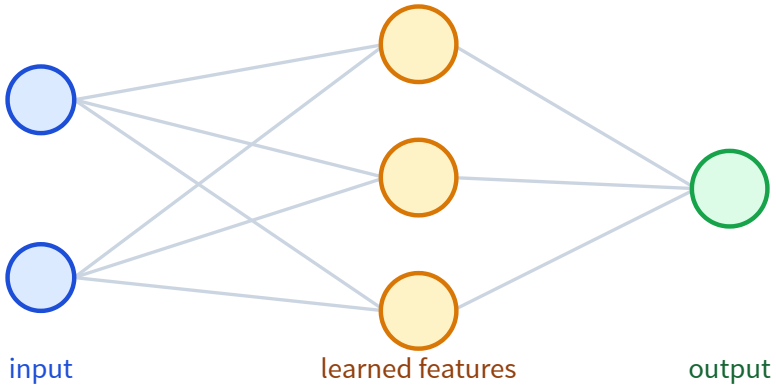
Train



activation: **OFF (linear)**
untrained — press **Train**

Same net, same data. Activation OFF: stacked linear layers stay a line. ON: the layer bends to fit the wave. That bend is what nonlinearity buys.

One hidden layer = learned feature map

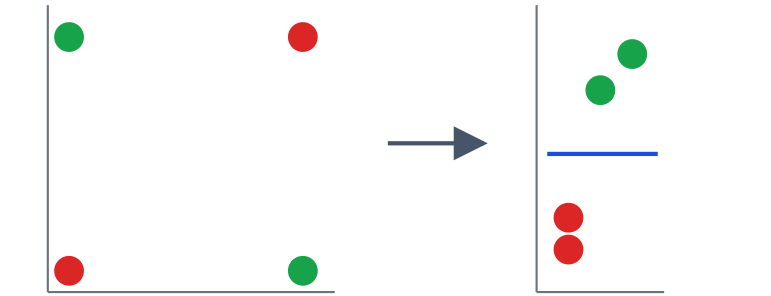


Each hidden unit learns a detector.

The output layer
combines detectors.

This is how fixed features
become learned features.

XOR: why hidden features help



raw input: not linearly separable

hidden features make it easy

A hidden layer can transform the space.

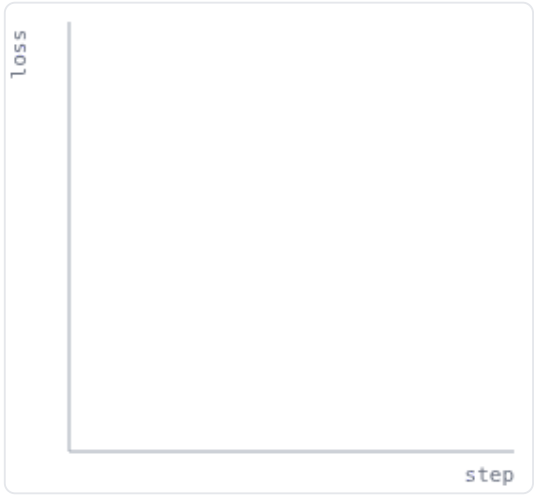
Then a simple output layer can solve the task.

Playground: train a net on XOR LIVE

Train a tiny net on XOR — try it on the live slides.

`nn.Sequential(nn.Linear(2,2), nn.Tanh(), nn.Linear(2,1))` · task: XOR

Train | activation: OFF



activation: **OFF**
untrained — press **Train**
○ class 0 ■ class 1

Green/red = the net's decision surface; markers are the true labels. Turn the activation OFF and the 2-2-1 net becomes linear — it cannot separate XOR no matter how long it trains.

An MLP: a bigger model, same loop

$$h = g(W_1 x + b_1)$$

$$\hat{y} = \mathrm{softmax}(W_2 h + b_2)$$

Every operation is differentiable, so L17's training loop applies unchanged.

Quick check: what changed from logistic regression?

Only the function class, not the optimization recipe.

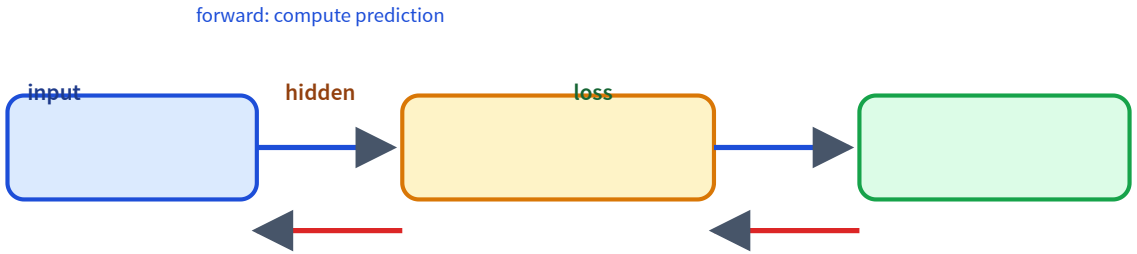
The code barely changes

```
model = nn.Sequential(
    nn.Linear(d_in, 64),
    nn.ReLU(),
    nn.Linear(64, n_classes),
)

loss = loss_fn(model(x), y)
loss.backward()
optimizer.step()
```

The function is richer; the training pattern is the same.

Backpropagation assigns credit



Mechanically, this is the chain rule applied through the computation graph.

What do hidden layers learn?

Deep models compose simple features into useful abstractions.

CNNs learn visual features

shallow: curves

middle: parts

deep: objects

one deep neuron = a "ball" detector

Feature visualizations (what each neuron responds to most).
Olah et al., "Feature Visualization," Distill 2017 (CC-BY).

Local filters detect small patterns.

Shallow: edges & curves.

Deeper: object parts.

Deepest: whole objects.

Later: VLMs reuse
image encoders (L23).

Act 2: embeddings give symbols geometry

The same representation idea applies
when the input is not pixels, but tokens.

A word ID is just a label; an embedding is a learned location in a vector space.

Word IDs have no geometry

IDs are arbitrary labels:

```
cat = 17, the
= 18, dog = 932
```

id 17 (cat) is no closer to id
18 (the) than to id 932 (dog)
— adjacency means nothing.

An embedding gives each a vector,
where **distance = similarity**:

Similar meaning → nearby vectors (cat close to dog; "the" is off on its own).

An embedding table is a learnable lookup

It is just a **lookup table** (a dictionary): a token's **id** selects its **row**.

token	id	row = learned vector
cat	17	[0.22, -0.71, 0.10, ...]
dog	932	[0.19, -0.66, 0.12, ...]
car	40	[-0.80, 0.31, -0.40, ...]

`emb[17]` → the "cat" row

```
emb = nn.Embedding(
    num_embeddings=vocab_size,
    embedding_dim=128,
)
```

Unlike a fixed dictionary, the rows are **parameters**: gradient descent updates them during training.

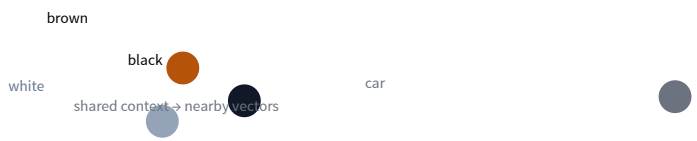
How does an embedding learn?

Words in the **same contexts** get pulled together:

`I have a ____ cat` is usually a color.

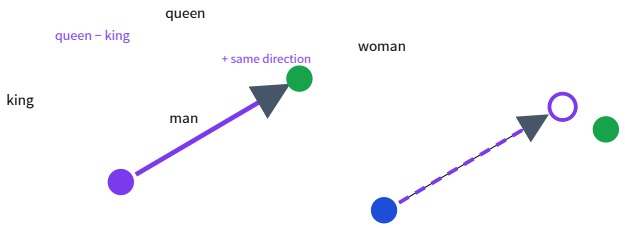
brown	black	white

To fill the same blanks, the model gives these color words **similar embeddings**.



Directions can carry meaning

king − man + woman ≈ queen



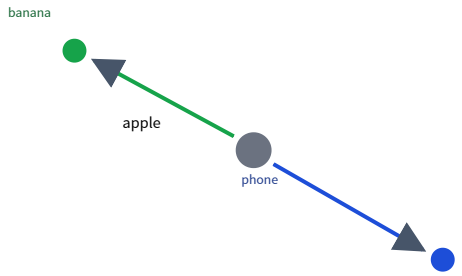
Add the "royal" direction to **man** and you land *near* woman (word2vec, GloVe).

Today this is learned **implicitly inside LLMs**;
contextual vectors rarely do clean arithmetic.

One word can mean two things

"I ate an **apple**" → fruit

"**Apple** released a phone" → company



A single fixed vector for "apple" cannot be both.

Modern models embed the **whole sentence**, so a word gets a different **contextual** vector each time.

This creates the L19 problem: context has to move information between token vectors.

Act 3: inspect learned spaces

Once inputs become vectors, we can ask geometric questions.

project

Can I see the space?

neighbors

What is close to what?

cluster

What groups exist?

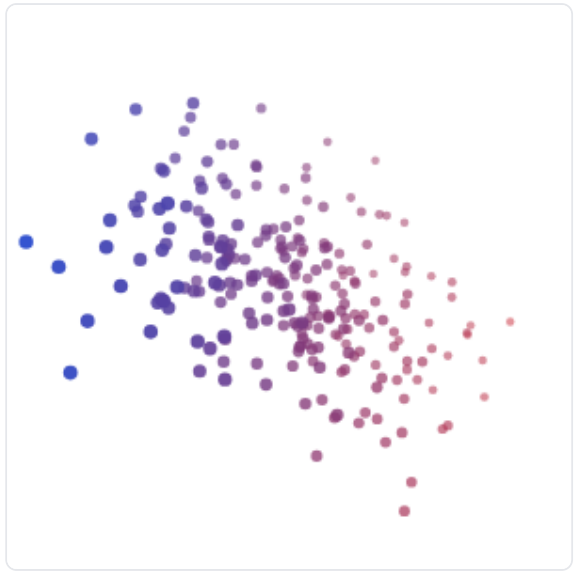
Inspect embeddings by projecting them LIVE

PCA keeps the directions of most spread — drag the cloud, then flatten it.

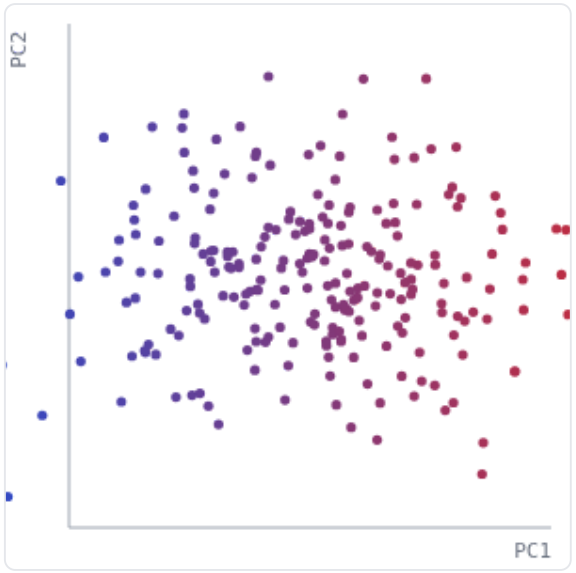
Drag the 3D cloud to rotate. PCA finds the flat plane of most spread and drops the thin 3rd direction.

Flatten to 2D Reseed

3D data (drag to rotate)



PCA → 2D



PCA on real data: a million chats LIVE DATA

Real WildChat conversations (1536-dim) projected to 2D — hover to read, click to open.

Each dot is a real conversation — a 1536-dim vector (OpenAI text-embedding-3-small) projected to 2D with PCA.

Zoom + Zoom − Reset view Explore it live at wildvisualizer.com ↗

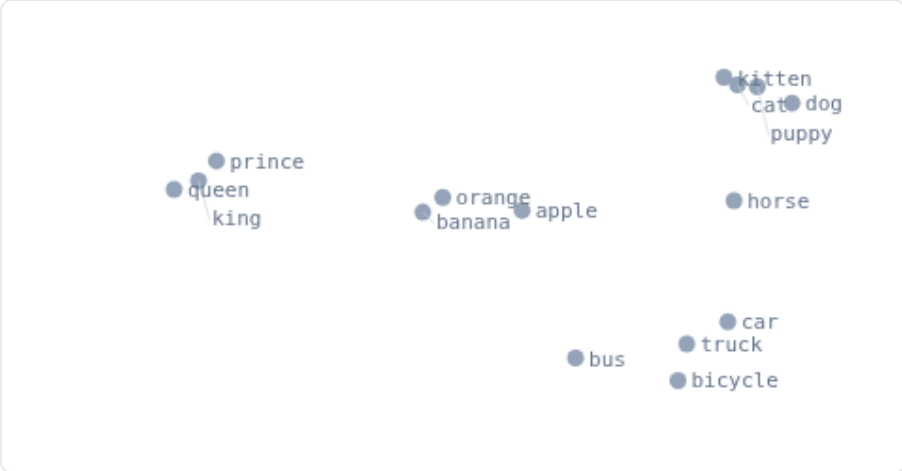
1428 real conversations — hover to read, click to open.

Playground: embed real words LIVE MODEL

Type words or sentences; a real model embeds them — try it on the live slides.

```
dog, puppy, cat, kitten, horse, car, truck, bus, bicycle, apple, banana,
orange, king, queen, prince
```

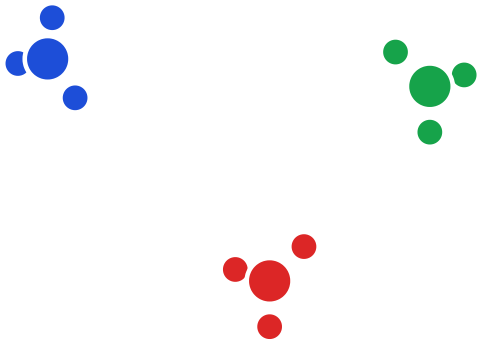
Embed Reset words



click a point to see its nearest neighbors

Type any words or short sentences. MiniLM computes a contextual embedding (not a fixed lookup), then PCA squeezes it to 2D. Similar meanings land near each other.

Cluster embeddings with k-means

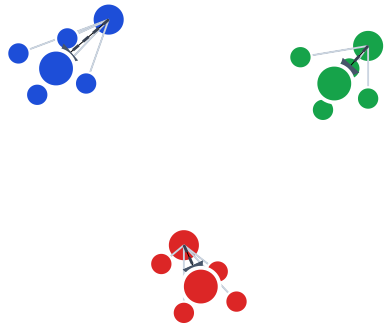


After embeddings exist, k-means becomes a way to summarize the space with prototypes.

- inspect datasets
- find groups or duplicates
- build codebook intuition: replace many vectors with a few representative centers

The k-means algorithm

1. Choose k; initialize k centroids (e.g. random data points).
2. Repeat until assignments stop changing:
    a. **Assign** — put each point with its nearest centroid.
    b. **Update** — move each centroid to the mean of its points.

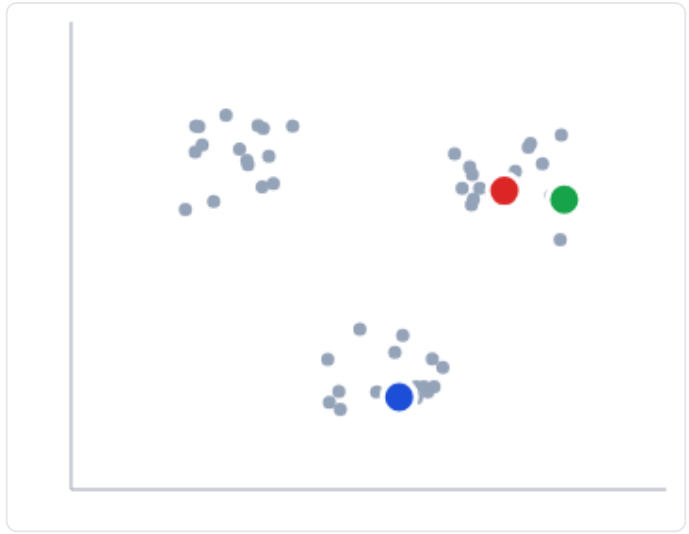


Converges to a **local** optimum — depends on initialization, so run a few times and keep the best.

Playground: k-means, step by step LIVE

Step through assign and update — try it on the live slides.

Step Run Reseed



```
iteration 0
centroids placed
```

Each step: assign every point to its nearest centroid, then move each centroid to its members' mean. Repeat until nothing moves.

Act 4: learned spaces become interfaces

So far, we used embeddings for individual points, neighbors, and clusters.

Next: use learned spaces to compare distributions, compress data, and align modalities.

How do you grade an image generator?

Prompt `"a dog"` — every sample is a **different** valid dog.

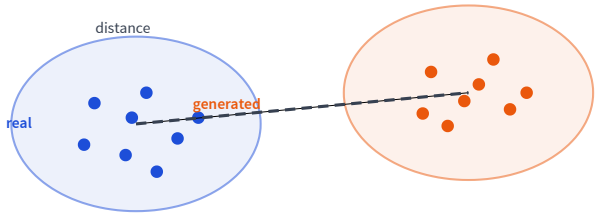
dog A	dog B	dog C
fluffy, sitting	running, brown	puppy, close-up

There is no single ground-truth image to compare against.

So compare the whole **distribution** of outputs to real images.

Measuring image quality: FID

Embed real and generated images with an Inception network, then compare the two **distributions** in feature space.



FID = distance between the two feature distributions
— lower means the generations look more real.

Heusel et al. 2017.

How do you grade a story generator?

Ask for a short story — there are **countless** good ones.

story A

a lost dog
finds home

story B

a robot learns
to paint

story C

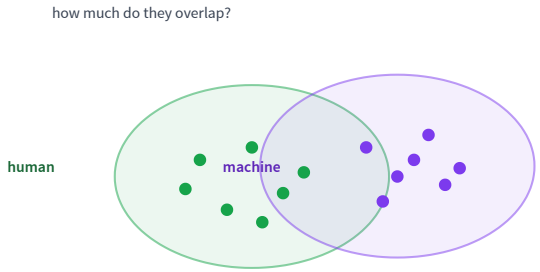
two friends
and a map

No ground-truth story to score against.

So compare machine stories to **human** stories as distributions.

Measuring text quality: MAUVE

Embed human and machine text in a language model's space and compare the two distributions.

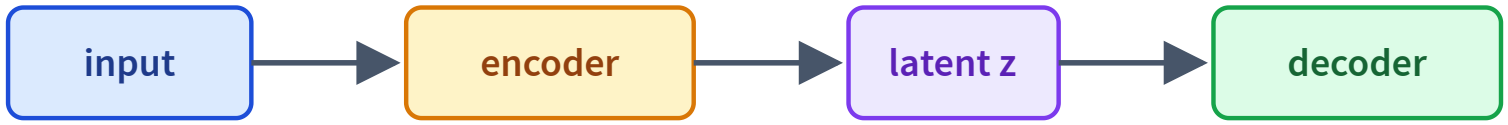


MAUVE summarizes how much the two distributions overlap — higher means more human-like.

Pillutla et al. 2021.

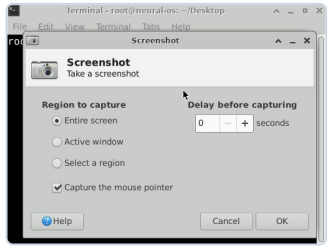
Autoencoders learn compression

Another use of learned spaces: store what matters in a smaller latent vector.



Latent diffusion will generate in a compressed latent space.

A real autoencoder: NeuralOS

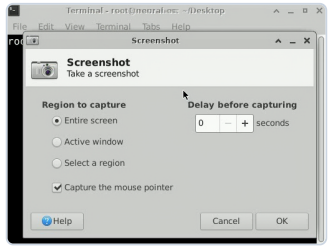


input

→ encoder →

latent z

→ decoder →



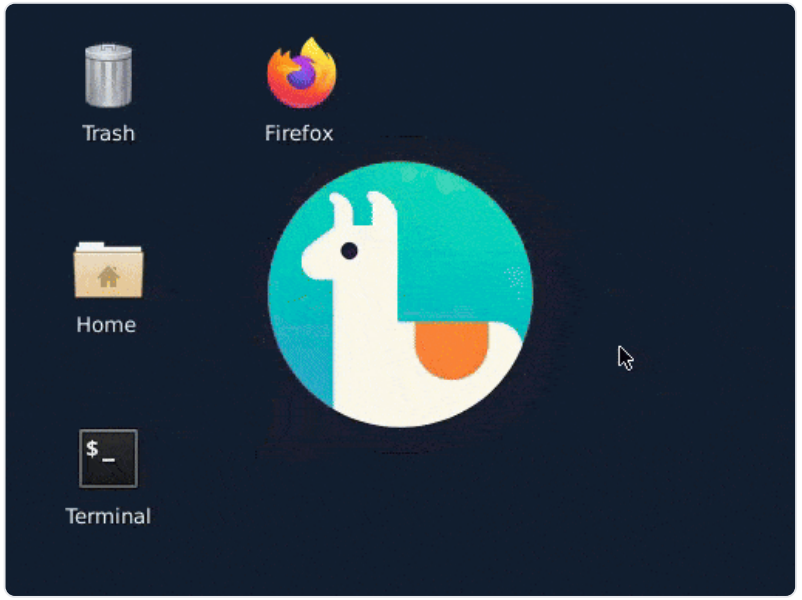
reconstruction

The decode looks the same as the input —
the small latent kept everything that matters.

This makes the latent a useful interface: compact enough to model, rich enough to reconstruct.

Autoencoder from NeuralOS (neural-os.com).

Why compress? To make generation feasible



neural-os.com

Generating in pixel space is expensive.

$$512 \times 384 = 196{,}608 \text{ pixels}$$
$$\rightarrow 64 \times 48 = 3{,}072 \text{ pixels}$$

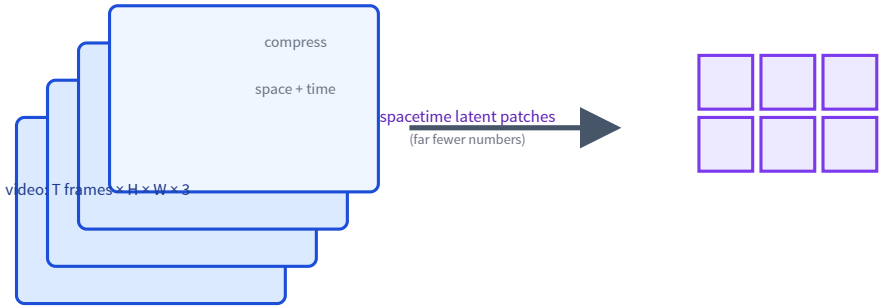
≈ 64× fewer pixels to generate.

NeuralOS predicts the next frame in this small latent — a neural operating system.

Same story as embeddings: do hard work in the learned space, not the raw space.

Video: compress space and time

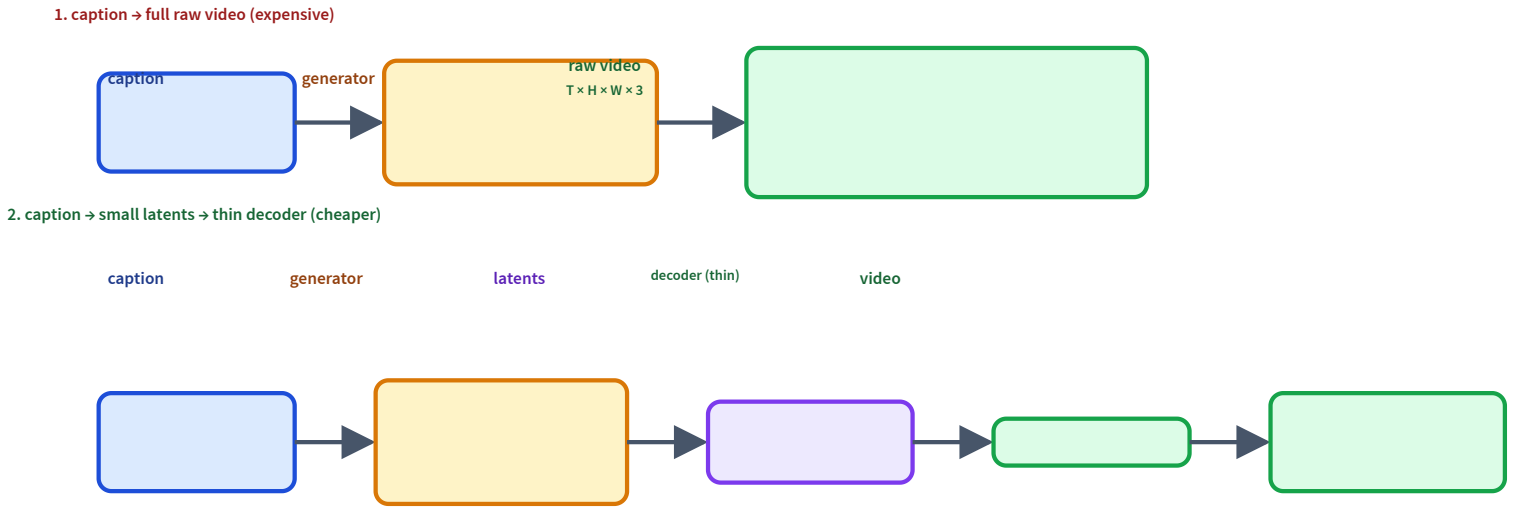
A video is many frames, so pixel cost explodes — compress the time axis too.



Sora compresses space and time into spacetime latent patches, then generates there.

OpenAI, "Video generation models as world simulators," 2024.

Two ways to generate video

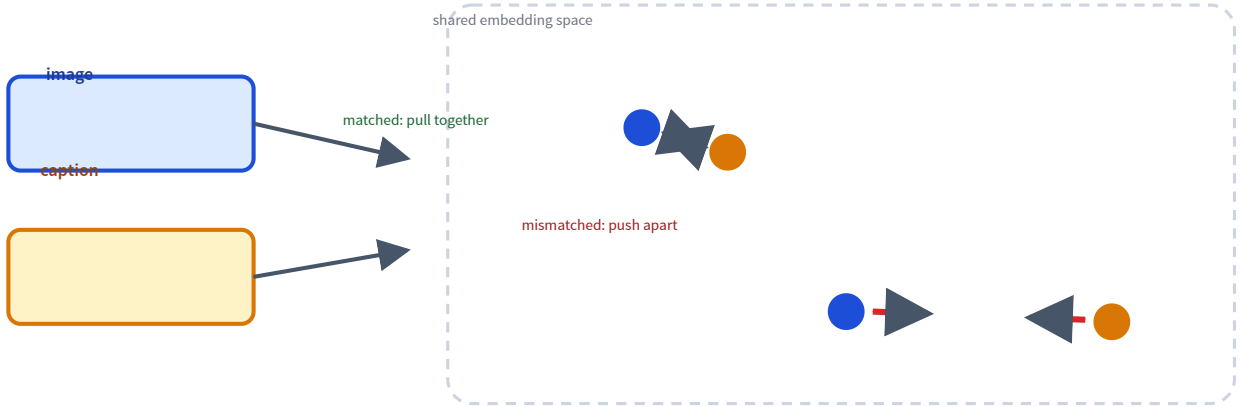


Model the small latents, not the pixels; a thin decoder maps them back to video.

Contrastive learning aligns modalities

Put images and their captions into one shared vector space.

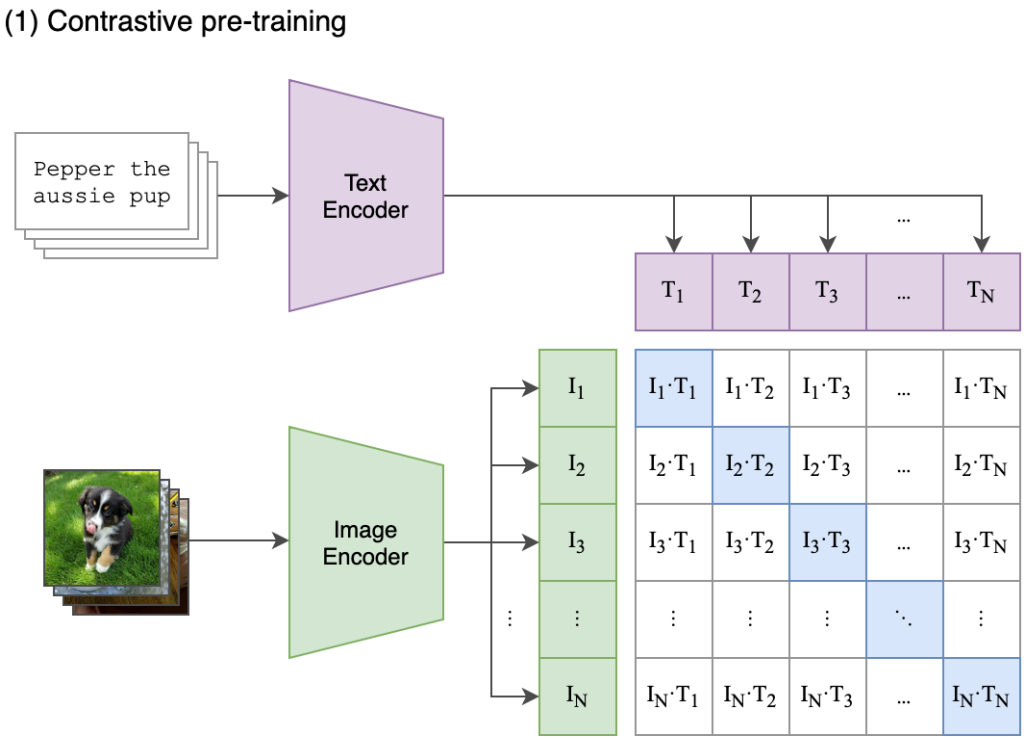
A final interface: two modalities become comparable by distance.



A dog photo lands near "a dog" and a car photo near "a car" — but the dog photo stays far from "a car".

CLIP: contrastive pre-training

In each batch, every image is compared with every caption; the correct pairs are the diagonal.



Across a batch, the matching image-text pairs (the diagonal) are trained to score highest.

Radford et al., "Learning Transferable Visual Models From Natural Language Supervision," ICML 2021.

What a shared image-text space unlocks

zero-shot classification

score an image against prompts like `"a photo of a {label}"` — no task-specific training

retrieval

search images with text (and vice versa) by nearest vectors in the shared space

This shared space is the bridge to vision-language models (L23).

Why this matters for LLMs

Returning to language: L18 gives the vector objects that L19 will move around.

tokens

become embeddings

attention

moves information between embeddings

transformers

scale this up

Carry this forward

From L17	L18 adds	Needed for L19
gradient training	hidden representations	deep sequence models
fixed feature vectors	learned embeddings	tokens as vectors
loss curves	PCA, neighbors, k-means	interpreting attention behavior

The through-line: learn a useful space, then compute inside it.

Recap

✓ Nonlinear hidden layers let neural nets learn **features**, not just predictions.

✓ Backprop assigns credit through the graph with the chain rule.

✓ **Embeddings** are learned vectors; contextual embeddings depend on the sentence.

✓ PCA, t-SNE/UMAP, neighbors, and k-means help inspect embedding spaces.

✓ FID, MAUVE, autoencoders, and CLIP show learned spaces acting as interfaces.

Next question

A sentence is a sequence of contextual embeddings.

How should each word look at the other words?

L19: sequence models, attention, and transformers.